B0310- Datasheet

Arducam 12MP IMX708 HDR 120°(H) Wide Angle Camera Module with M12 Lens for Raspberry Pi

SKU

B0310

- Complementary to RPI V3: An M12 lens supplement to the Raspberry Pi IMX708 autofocus camera module 3.
- Superior Sensor: Sony IMX708 with HDR mode for up to 3-megapixel output, MIPI CSI-2 Communication.
- Replaceable M12 Lens: Equipped with a 120°(H) wide-angle M12 Lens. You can replace it with other M12 lenses. Example Lenses: LK003
- Platform Compatibility: Supports Raspberry Pi 5, 3, 4 A/B, equipped with 15-22pin and 15-15pin FPC flexible cables.
- Cost-effective: second-to-none choice for given image and cost.



Overview

Raspberry Pi Camera Module 3 is the latest camera model released by Raspberry Pi Official in 2023. Arducam has designed two Fixed-Focus camera variants before. This camera adopts Sony 11.9MP IMX708 Image Sensor which can be used to take full HD video as well as still photographs and features an HDR mode of up to 3 megapixels. The camera is equipped with a 120° wide-angle M12 Lens. Wide-Angle Lens always has a deeper depth of field which can effectively reduce bokeh. A manual focus camera allows you to use it in situations such as macro photographs, low light, etc. where autofocus is difficult to achieve. Meanwhile, it is native to Raspberry Pi and fully supported by libcamera and Picamera2 library.
Arducam is committed to creating diverse and convenient camera solutions. Also, we designed various accessories for the Raspberry Pi Camera Module 3 which include the Camera Extension solution, Quad-Vision stereo camera kit, UVC USB camera solution, and upcoming FPD-Link solution, etc. For details, please refer to our official website www.arducam.com

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Applications

- Wildlife Photograph
- Machine Learning
- Home Security System
- AI applications
- DSLR Camera Control

Features

- Complementary for Official: An M12 Lens supplement to the Raspberry Pi IMX708 autofocus camera module 3, catering to manual focus favorable use cases.
- Arducam Camera Module 3 Series: Based on back-illuminated stacked CMOS 12-megapixel high-resolution Image Sensor (Sony IMX708) with HDR mode for up to 3-megapixel output on Raspberry Pi. Max Resolution: 4608 × 2592.
- Wide Angle M12 Lens: Armed with a 120°(HFOV) Wide Angle M12 Lens. A wide-angle lens has a deeper depth of field, which can effectively reduce bokeh. At the same time, they can better include foregrounds for landscape photos and improve image quality.

- Manual Focus: Manual Focus cameras are always more suitable in some special conditions. For example, in low-light and macro photography environments, manual focus can obtain a more complete overall contrast of pictures, thereby improving image quality and focus accuracy. In HDR mode, panorama stitching, and other modes, manual focus can decrease different focus layers and blur efficiently.
- Platform Compatibility: Compatible with Raspberry Pi 3, 4 A/B, equipped with 15-22pin and 22-22pin FPC flexible cables. Support used on Raspberry Pi Zero.

Specifications

Camera

Image Sensor	IMX708
Still Resolution	11.9 Megapixels
Video Modes	2304 × 1296p56, 2304 × 1296p30 HDR, 1536 × 864p120
Sensor Resolution	4608 x 2592 pixels
Sensor image area	6.45 x 3.63mm (7.4mm diagonal)
Pixel Size	1.4µm × 1.4µm
Optical Size	1/2.43"
IR Sensitivity	Integral IR-cut Filter, visible light only
Supported Platform	Raspberry Pi 5/4B/3B+/3/2/CM3/CM4/CM5/Zero W/Zero 2 W

Lens

Focus Type	Manual Focus

Focal Length	2.87mm
F.NO	F2.8
Field of View(FOV)	152°(D)x120°(H)x66°(V)
Lens Mount	M12

Electrical and Mechanical

Camera Board Size	25 × 24mm